

Simulation of a gamma spectrum acquired with a sodium iodide scintillation detector - NaI(Tl) -

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1 Introduction

One application of inorganic NaI(Tl) scintillation detectors is the monitoring of environmental radioactivity. A radioactive source may decay by emitting gamma radiation. A gamma photon emitted can interact with the detector, depositing some (or all) of its energy. The energy deposited by the photon excites the atoms in the scintillator, which emit light upon de-excitation. This light is converted into an electrical signal by a photodiode; as there is a proportional relation between the intensity of the generated electrical signal and the fluorescence light, and between the intensity of the fluorescence light and the energy of the gamma photon, the detector is able to estimate the energy of the gamma photon and thus identify the radioactive source. A photon entering the detector can deposit energy by being absorbed via the photoelectric effect (all of the initial energy is absorbed by the detector) or via the Compton effect (only part of the energy is deposited in the detector). However, there are certain factors that can limit the information content of a gamma-ray spectrum. The first aspect is the detector's resolution, that is, the measure of how faithfully it converts the energy deposited by the photon into an electrical signal. The second factor is the following: a photon that undergoes one or more Compton scatterings and exits the detector with residual energy has deposited a quantity of energy in the detector that is less than its initial energy. Such a photon will contribute to the gamma spectrum with a count at an energy different from its initial energy and because of that it will not "contribute effectively" to the identification of the radioactive source. The aim of this simulation is to recreate a gamma spectrum measured by a NaI(Tl) scintillation detector. A simulated spectrum can be used to support the analysis of closely spaced peaks (such as those at 609 and 662 keV of Bi-214 and Cs-137, which are relevant for environmental monitoring applications) in measured gamma spectra and to investigate aspects of the physics of interactions between photons and the detector. In the latter case, it is possible, for example, to evaluate the effect on the gamma spectrum of the relative probability that a photon, depending on its energy, will undergo Compton scattering or the photoelectric effect.

2 Source activity

The initial number of atoms N_0 in a radioactive source is given by:

$$N_0 = \frac{m}{M} N_a \quad (1)$$

where m is the mass of the source, M is the molar mass and N_a is the Avogadro number. The number of atoms N , that will *not* undergo radioactive decomposition in a time t is given by:

$$N = N_0 e^{-\lambda t} \quad (2)$$

where λ is the *decay constant* of the source. We can also define the *half-life* ($t_{1/2}$) as the time needed to reduce the number of the radioactive nuclides to half; $t_{1/2}$ is related to the decay constant by the relation:

$$\lambda = \frac{\ln(2)}{t_{1/2}} \quad (3)$$

Finally, we define the *radioactivity* or *absolute activity* (A) of the source as the number of atoms decaying in the unit time; of course the activity is proportional to the number of initial nuclei N_0 :

$$A = \frac{dN_0}{dt} = -\frac{d}{dt} N e^{\lambda t} = \lambda N e^{\lambda t} = \lambda N_0 = \frac{\ln(2)}{t_{1/2}} \frac{m}{M} N_a \quad (4)$$

To get the gamma intensity, the activity must be corrected considering that gamma emission follows the relaxation of a first decay to a meta-stable state; in the case of Cs-137, the direct decay to the stable state of Ba-137 has probability 5.6% to occur, while the probability of decay to the metastable state linked to the 662keV gamma emission is 94.4%. For such reaction, at a given activity A , the gamma intensity I_γ is:

$$I_\gamma = 0.944 \times A \quad (5)$$

3 Gamma-environment interaction

The scenario we depict is that of a point radioactive source located in the ground at a certain depth; the detector is positioned on the ground. The gamma intensity I reaching the detector is given by:

$$I = I_\gamma \frac{r_d^2}{4\pi R^2} e^{-\lambda_s R} \quad (6)$$

In which r_d^2 is the detector base area (in our case, $10\text{cm} \times 10\text{cm}$), R is the source-detector distance, A is the source activity, I_γ is the intensity of the emitted gamma and λ_s is the linear attenuation coefficient of the soil, which depends on the soil density ρ_s and the soil mass attenuation coefficient μ_s :

$$\lambda_s = \mu_s \rho_s \quad (7)$$

4 Gamma-detector interaction

4 Photoelectric effect

In a photoelectric effect, a photon transfers his energy to a free or bounded electron in the material, resulting in the latter case (if the energy is sufficient), in the emission of the electron from its bounded state. The remaining energy (or all the initial energy if the electron was free) is transferred as kinetic energy of the electron.

$$E_k = E_\gamma - E_b \quad (8)$$

Where E_k is the kinetic energy transferred to the electron, E_γ is the energy of the gamma photon and E_b is the binding energy; in the simulation, the approximation we made, is that in a photoelectric event, the photon transfers all its energy as kinetic energy of the electron ignoring the binding energy (in other words, all the electrons are considered as free). This means that all the photon energy is deposited in the material when photoelectric effect happens.

4 Compton scattering

The Compton effect, is an inelastic scattering phenomenon between a photon and a charged particle (an electron). The photon, in our case a γ photon, transfers part of its energy to the electron and it's scattered at an angle θ . The energy transferred to the electron is given by:

$$E_e = E_\gamma \left[1 - \frac{1}{1 + \frac{E_\gamma(1-\cos(\theta))}{m_0c^2}} \right] \quad (9)$$

Where E_e is the electron energy, E_γ is the initial energy of the photon, m_0 is the rest mass of the electron ($m_0c^2 = 511keV$) and θ is the scattering angle; For a $662keV$ photon, the maximum transferable energy is $477keV$ (when the photon is backscattered).

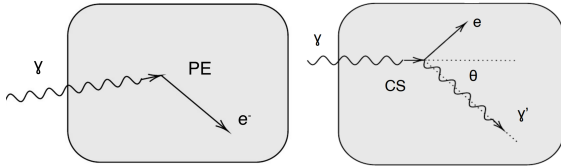


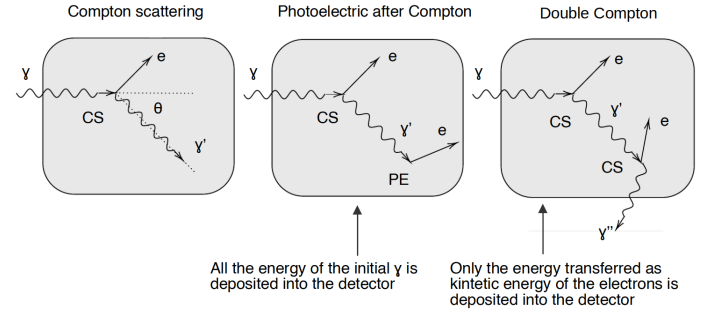
Figure 1: Photoelectric effect and Compton scattering

The probability for a photon to be scattered at a given angle, is determined by the Klein-Nishina equation. The energy of the photon after the Compton scattering ($E_{\gamma'}$) is given by the difference between the initial energy of the photon E_γ and the energy transferred to the electron:

$$E_{\gamma'} = E_\gamma - E_e \quad (10)$$

As mentioned in the introduction, a photon is able to deposit all its energy in the detector only if at some point it undergo

photoelectric effect; if the photon exits the detector after zero or more Compton scattering, the energy will be the energy deposited via Compton scattering(s) up to that point



5 CS-PE cross section

The probability of a photon being absorbed via the photoelectric effect or losing energy through Compton scattering is defined by the 'crossSectionRatio' function (CSR), which represents the ratio between the probabilities of the photoelectric and Compton effects occurring as a function of energy. The probability of the photoelectric effect occurring decreases as energy increases, and this is implemented by constructing the CSR function as a simple negative exponential:

$$CSR = \beta \exp(-\alpha E) \quad (11)$$

In which E , is the energy of the interacting photon. This is, of course, a toy model, which can be analysed by estimating the α and β parameters or refined by introducing more accurate models.

6 Number of Compton scattering events

When a photon enters the detector, it may undergo one or more Compton scattering events before exiting the detector or being absorbed via the photoelectric effect. The maximum number of scattering events (N_{CS}) a photon can undergo before exiting the detector is calculated by sampling from a negative exponential function:

$$f(N_{CS}) = \exp(-\mu N_{CS}) + 1 \quad (12)$$

To summarise: the maximum number of Compton scattering events a photon can undergo before exiting the detector is determined by the function $f(N_{CS})$; however, after each scattering event, the CSR function calculates the probability for the photon to be completely absorbed via photoelectric effect as a function of its remaining energy E up to that point, and consequently determines the actual number of Compton scatterings (up to a maximum set by $f(N_{CS})$) for each photon.

7 The Klein-Nishina distribution

For an incident non-polarized photon, we define the Compton scattering differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{1}{2} r_e^2 \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2(\theta) \right] \quad (13)$$

where r_e is the classical electron radius (~ 2.82 fm), ω'/ω is the ratio of the angular frequencies of the incident and the scattered photons and θ is the scattering angle. The probability density function of the scattering angle (i.e. the Klein-Nishina distribution) is proportional to the scattering differential cross section:

$$f(\theta) \propto \frac{d\sigma}{d\Omega} \quad (14)$$

and can be found by imposing the normalisation condition:

$$\int \frac{d\sigma}{d\Omega} d\Omega = 1 \quad (15)$$

In which the solid angle $d\Omega$ is:

$$d\Omega = \sin(\theta) d\theta d\phi \quad (16)$$

And therefore:

$$\int \frac{d\sigma}{d\Omega} d\Omega = \int \frac{d\sigma}{d\Omega} \sin(\theta) d\theta d\phi = \int_0^\pi \frac{d\sigma}{d\Omega} \sin(\theta) d\theta \int_0^{2\pi} d\phi = 1 \quad (17)$$

The cross section as a function of the angle ϕ is expected to be constant; we define the function $I(\theta)$ as:

$$I(\pi) - I(0) = \int_0^\pi \frac{d\sigma}{d\Omega} \sin(\theta) d\theta \quad (18)$$

Such integral has solution:

$$\begin{aligned} I(\theta) \propto \int d\theta \sin(\theta) \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2(\theta) \right] = \\ -\frac{\cos(\theta)}{R^2} + \ln\{1 + R[1 - \cos(\theta)]\} \left(\frac{1}{R} - \frac{2}{R^2} - \frac{2}{R^3} \right) - \\ \frac{1}{2R\{1 + R[1 - \cos(\theta)]\}^2} + \frac{1}{1 + R[1 - \cos(\theta)]} \left(-\frac{2}{R^2} - \frac{1}{R^3} \right) \end{aligned} \quad (19)$$

In which:

$$R = \frac{\hbar\omega}{m_0 c^2} \quad (20)$$

The normalisation condition become:

$$\int_0^\pi d\theta \frac{d\sigma}{d\Omega} \sin(\theta) \frac{1}{I(\pi) - I(0)} = \int_0^\pi d\theta f(\theta) = 1 \quad (21)$$

But the $I(\theta)$ function, si related to the cumulative distribution function (CDF) of the scattering angle θ ; infact, by the definition of the CDF:

$$\begin{aligned} F(\theta) &= \int_0^\pi d\theta f(\theta) \\ &= \int_0^\theta d\theta \frac{d\sigma}{d\Omega} \sin(\theta) \frac{1}{I(\pi) - I(0)} = \frac{I(\theta) - I(0)}{I(\pi) - I(0)} \end{aligned} \quad (22)$$

Hence, it's finally possible to fine the $f(\theta)$ function, i.e. the probability density function of the scattering angle:

$$f(\theta) = \frac{dF(\theta)}{d\theta} = \frac{1}{I(\pi) - I(0)} \sin(\theta) \left(\frac{\omega'}{\omega} \right)^2 \left[\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2(\theta) \right] \quad (23)$$

In the code, the Klein-Nishina function ("knpdf" in the code) is implemented by defining the functions $I(\theta)$, $R(\omega)$, $\omega(E)$ and $\omega'(E)$; once the function had been defined, it was possible to sample from it by implementing a rejection sampling algorithm. The number of samples is equal to the number of photons that initially undergo Compton scattering.

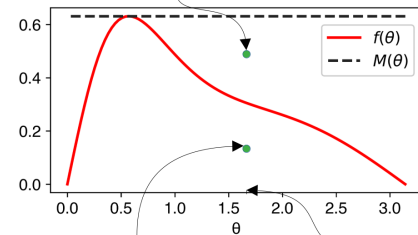
8 The rejection sampling algorithm

We want to sample from the Klein-Nishina probability density function $f(\theta)$. We define a function $M(\theta)$ (from which we can "easily sample"), which is defined in the same domain as $f(\theta)$ and such that $M(\theta) \geq f(\theta) \forall \theta$; the easiest choice for $M(\theta)$ is a uniform distribution:

$$M(\theta) = \max\{f(\theta)\} \quad (24)$$

The rejection sampling is structured in two steps; firstly a random θ_0 value is sampled with a uniform probability in the interval $[0; \pi]$ and $M(\theta_0)$ is computed as well as $f(\theta_0)$. Then, a random sample u is generated with a uniform probability in the interval $[0; M(\theta_0)]$, which in our case, is always $[0; \max\{f(\theta)\}]$; if $u \leq f(\theta_0)$, θ_0 is accepted as a sample of the random variable θ , otherwise is rejected:

a $u(\theta_0)$ value that would be rejected



a $u(\theta_0)$ value that would be accepted θ_0 value

Figure 2: The rejection algorithm

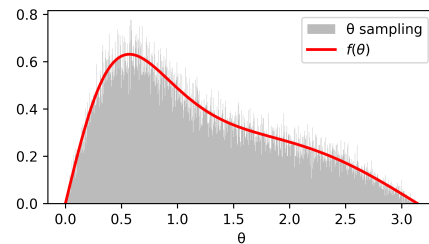


Figure 3: An example of θ sampling

For each scattering event, a sample of the Klein-Nishina distribution is generated using the rejection algorithm, and this sample is used to calculate the photon's energy loss according to equation 9.

9 Energy resolution

The detector resolution as a function of energy is:

$$\text{Res} = 1.714824 \cdot E^{-0.504} \quad (25)$$

The measure of the energy deposited by each photon, is assumed to be normally distributed around its true value; the full width at half maximum (FWHM) and the variance σ of such normal distribution are related to the detector resolution:

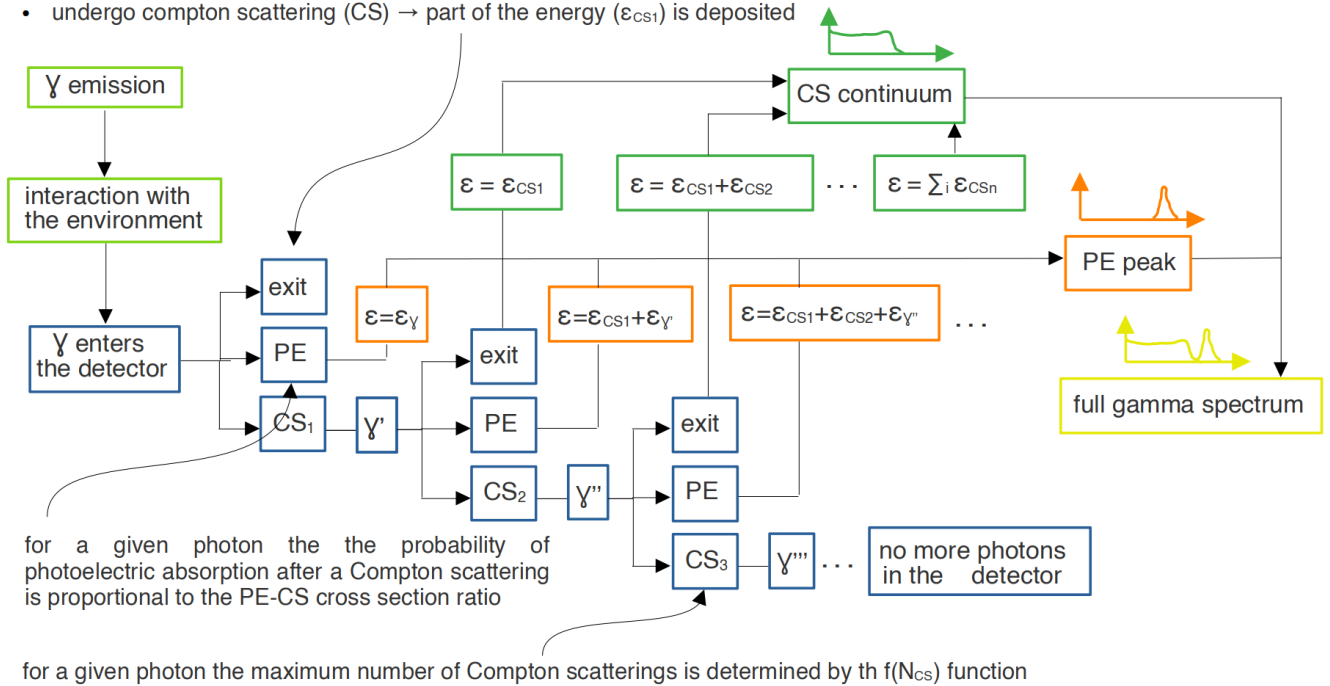
$$\text{FWHM} = \text{Res} \cdot E; \quad \sigma = \text{FWHM}/2.355 \quad (26)$$

In the simulation, the expected value of the deposited energy is the one given by equation 9 in the case of Compton scattering and it's the total initial energy of the photon in the case of photoelectric absorption. The simulated measured deposited energy is therefore sampled from a gaussian distribution with mean value equal to the expected value of the deposited energy and variance equal to σ (computed with equation 26), which depends on the expected value of the deposited energy itself, according to the relations 26. In the section below the flowchart of the simulation is shown.

10 Simulation flowchart

when a photon enters the detector, it can:

- exit right away $\rightarrow$ no energy deposited
- undergo photoelectric effect (PE) $\rightarrow$ all the energy (ϵ_γ) is deposited
- undergo compton scattering (CS) $\rightarrow$ part of the energy (ϵ_{CS1}) is deposited



11 Case study, Caesium-137

In the example shown in the code, a gamma spectrum for Cs-137 is generated. A Cs-137 source with a mass of 1.5 g (corresponding to a total activity of approximately 4.8 kBq) is assumed to be placed in the soil at a distance of 0.1 m from the detector (which is placed on the ground). The soil density and soil mass attenuation coefficient are assumed to be respectively $\rho_s = 1.3$ and $\mu_s = 0.08045 \text{ cm}^2/\text{g}$; the side length of the detector is 0.102m; Cs-137 has an energy peak of 661.7 keV and a Compton edge of 477.37 keV (corresponding to the backscattering of 661.7keV photons). The parameters α and β of the CSR function are taken as $\alpha = 0.001$ and $\beta = 0.05$. The parameter μ of the $f(N_{CS})$ function is taken to be $\mu = 3$. The acquisition time is set to 2 minutes. The photons that initially undergo compton scattering are 46413;

the photons absorbed immediately by photoelectric effect are 1229; the photons absorbed by photoelectric effect after one or more Compton are 5188; below is the spectrum, also shown on a logarithmic scale.

